STRUCTURAL · MODAL · VIBRATION QUALIFICATION ANALYSIS

MIRROR MOUNT ASSEMBLY

Modal & Random Vibration Analysis Report

Finite Element Analysis performed in ANSYS Mechanical 2025 R2

Assembly	Mirror Mount (Optical Sub-Assembly)
Document Type	Analysis Report
Analyses Covered	Modal · Random Vibration (PSD)
Solver / Software	ANSYS Mechanical 2025 R2
Report Date	17 July 2026
Status	For Review

Prepared for internal engineering review. Distribution outside the project team requires approval.

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Table of Contents

List of Figures

List of Tables

List of Symbols, Abbreviations & Units

Symbols and Quantities

Symbol / Term	Description	Unit
f	Natural (modal) frequency	Hz
σ	Standard deviation of a random response (statistical level)	1σ, 2σ, 3σ
µrad	Micro-radian — unit of mirror tilt / rotation	10^{-6} rad
PSD	Power Spectral Density	$(m/s^2)^2/Hz$
RMS	Root Mean Square value (1σ statistical response)	—
UY	Out-of-plane nodal displacement (local Y)	m
LOCX	Local radial coordinate of a node	m
θ	Small-angle tilt / rotation of the mirror surface	rad
Eff. Mass	Modal effective mass participation factor	kg / %
Max / Min	Maximum / minimum reported field value	—
MPa / kPa / Pa	Stress or pressure units (mega- / kilo- / base Pascal)	N/m^2
3σ	Three-sigma probability level (99.73%)	—

Abbreviations

Abbreviation	Meaning
CAD	Computer-Aided Design
FEA	Finite Element Analysis
FE	Finite Element
BC	Boundary Condition
DOF	Degree of Freedom
BOM	Bill of Materials
SS	Stainless Steel
AL	Aluminium Alloy

1. Introduction

1.1 Purpose and Scope

This report presents the finite element analysis (FEA) performed on the Mirror Mount assembly to verify its structural and optical performance under the specified dynamic environment. The evaluation comprises two linked studies: (i) a modal (normal-modes) analysis to characterise the assembly's natural frequencies and mode shapes, and (ii) a random vibration (PSD) analysis to predict the statistical structural response and resulting mirror tilt under a qualification-level vibration environment.

The primary acceptance criterion for the assembly is optical: the mirror surface tilt induced by structural deformation must remain within the alignment budget of the optical system under the vibration load case.

1.2 Acceptance Criteria Summary

Overall Design Targets

- Modal: first (fundamental) mode frequency $f \geq 300$ Hz, with cumulative modal effective mass ≥ 80% captured within the analysed mode set.
- Random Vibration: mirror tilt ≤ 5 µrad at the 3σ (99.73% probability) response level under the specified input PSD.

Each of these criteria is addressed in the corresponding section of this report, together with the analysis methodology, boundary conditions, and results.

2. Assembly Description and Finite Element Model

2.1 Assembly Overview

The mirror mount is a precision opto-mechanical sub-assembly comprising sixteen distinct components, dominated by the fused-silica mirror, a stainless-steel fixture mount, and an aluminium base plate / outer ring / flange structure that provides adjustable support through a micrometer and locking-pad arrangement. The complete assembly mass is 30.32 kg.

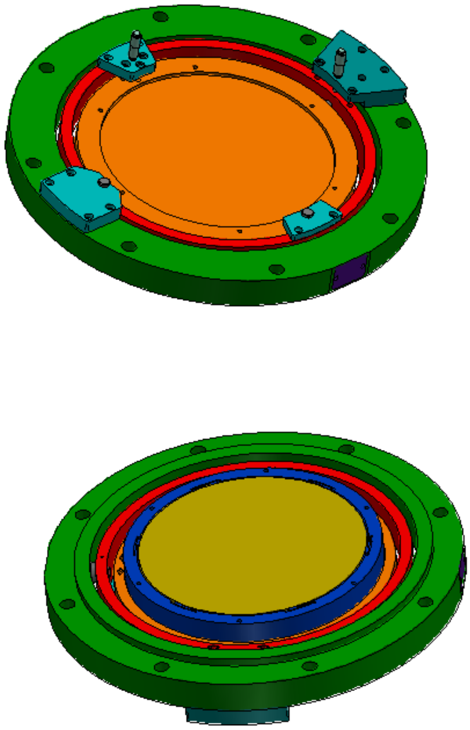


Figure 1: *Mirror mount assembly — isometric solid view.*

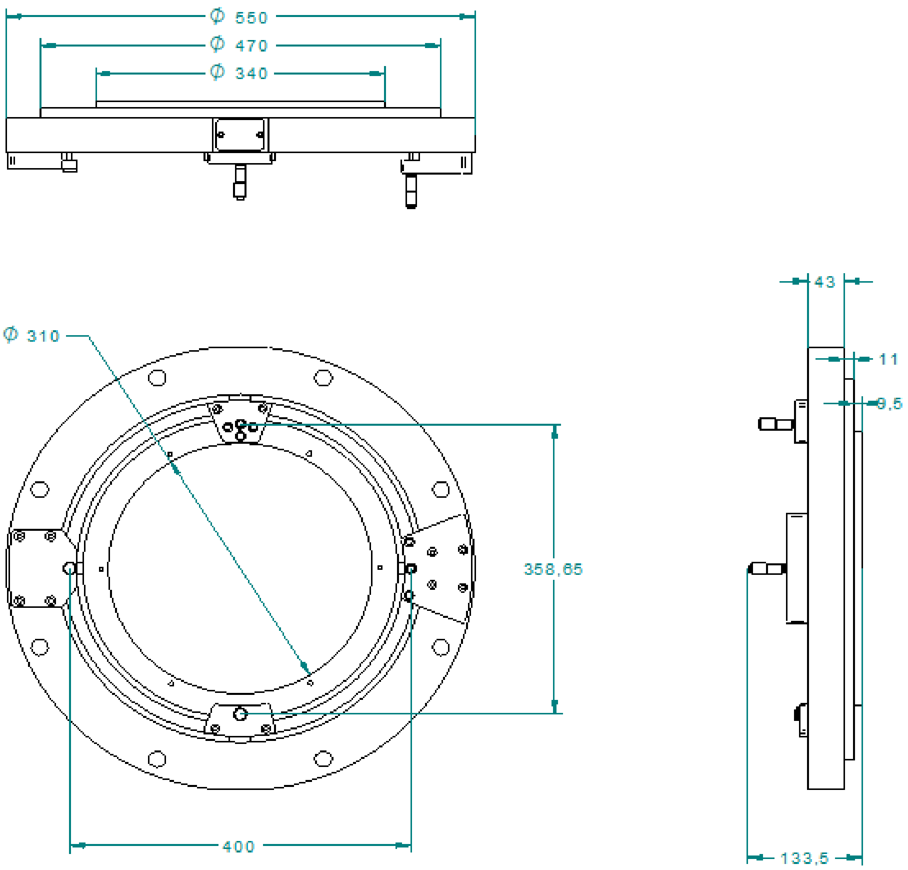


Figure 2*: Mirror mount assembly — sectioned / detail view.*

Table 1 lists the full Bill of Materials, cross-referenced by item number to the exploded view in Figure 3 directly below it, so that each row can be traced to its physical location in the assembly.

Table 1: *Bill of Materials for the mirror mount assembly.*

Item	Component	Material	Qty	Mass (kg)
1	MIRROR	Fused Silica	1	4.66
2	FIXTURE_MOUNT	SS 304	1	4.41
3	BASE_PLATE	AL 6061	1	3.92
4	OUTER_RING	AL 6061	1	2.32
5	FLANGE	AL 6061	1	10.49
6	OUTER_SHAFT	SS 304	2	0.168
7	INNER_SHAFT	SS 304	2	0.086
8	BEARING_CAP	SS 304	2	0.76
9	MICROMETER_MOUNTING_PAD_1	SS 304	1	1.53
10	MICROMETER_MOUNTING_PAD_2	SS 304	1	0.35
11	LOCKING_PAD_1	SS 304	1	0.976
12	LOCKING_PAD_2	SS 304	1	0.22
13	6202_1 (Bearing)	OEM	4	0.18
14	LOCK SCREW	OEM	2	0.01
15	BALL_1	SS 304	2	0.01
16	MICROMETER HEAD 25 MM_1	OEM	2	0.23
	TOTAL ASSEMBLY MASS			**30.32**

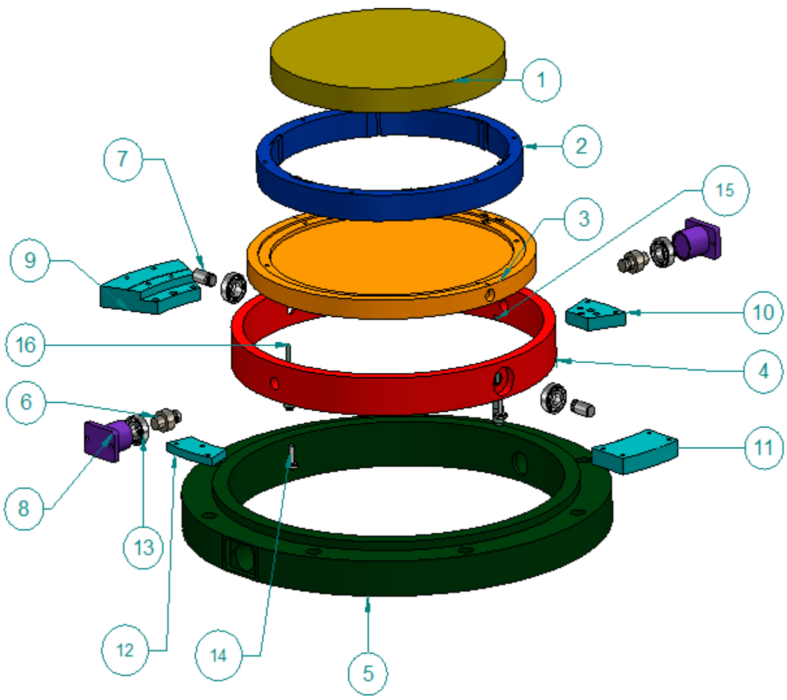


Figure 3: *Exploded view of the mirror mount assembly showing all sixteen components (item numbers per Table 1).*

2.2 Finite Element Mesh

The assembly geometry was imported, cleaned of non-essential features (fillets, threads, and other manufacturing detail not expected to influence global stiffness or mass), and meshed using a tetrahedral/hexahedral-dominant scheme appropriate to each component's geometry. Contacts between mating parts were defined per the contact schedule in Table 2, with all interfaces modelled as bonded, consistent with a permanently fastened / bonded assembly.

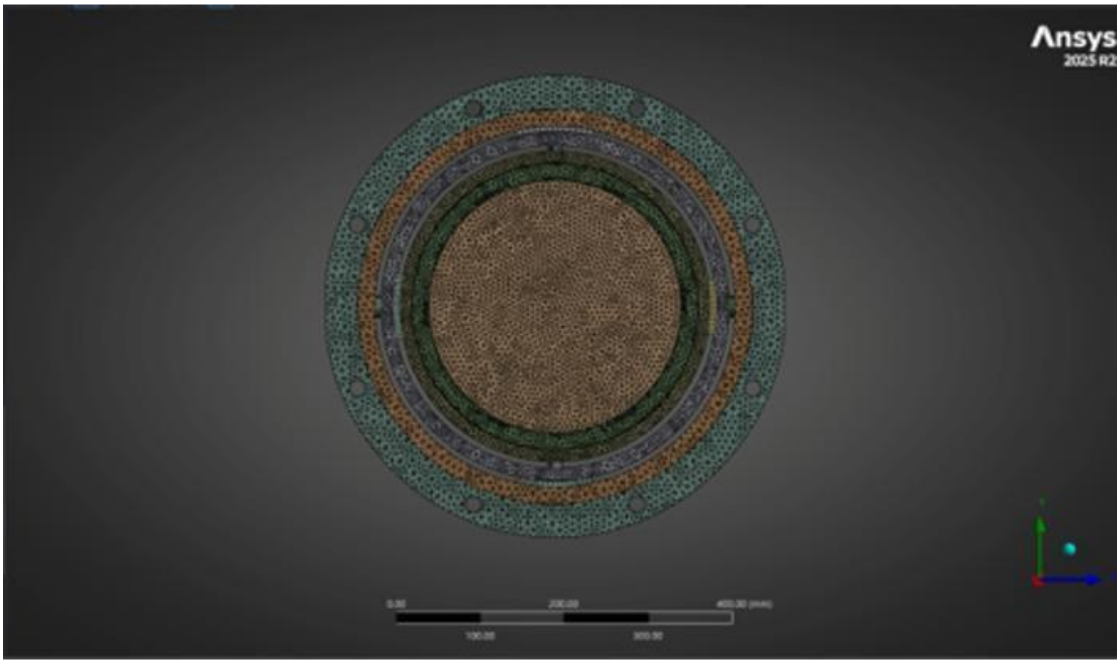


Figure 4: *Overall meshed assembly — isometric view.*

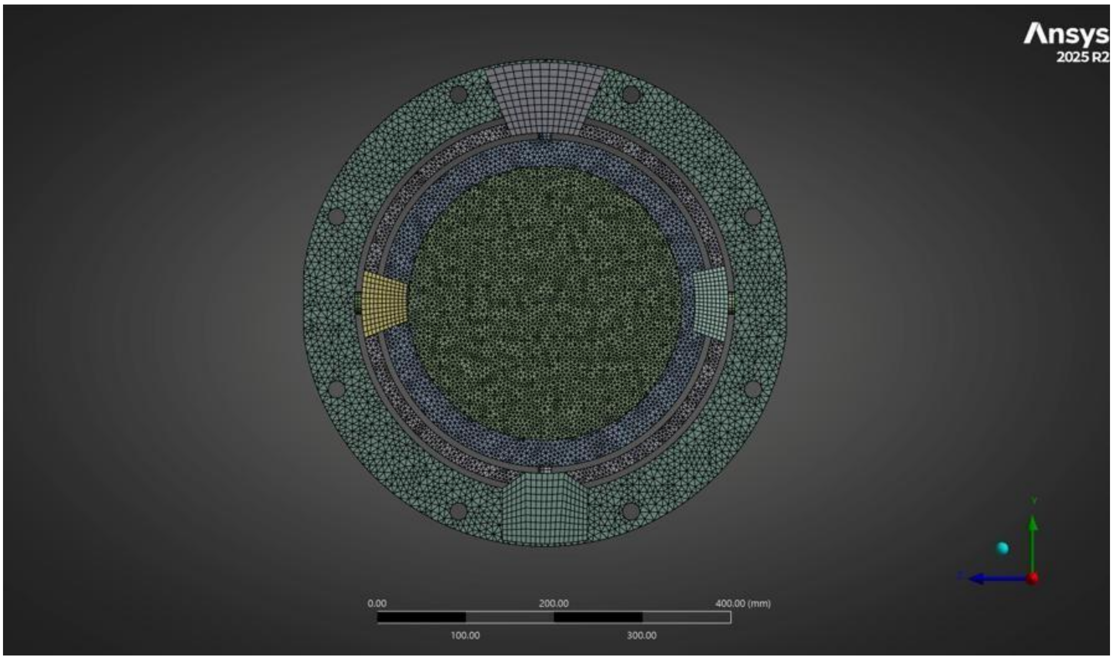


Figure 5: *Overall meshed assembly — alternate view.*

2.2.1 Meshed Components

Individual component meshes are shown in Figure 6 below; local mesh refinement was applied at fillets, bores, and contact regions to better resolve local stress and contact behaviour.

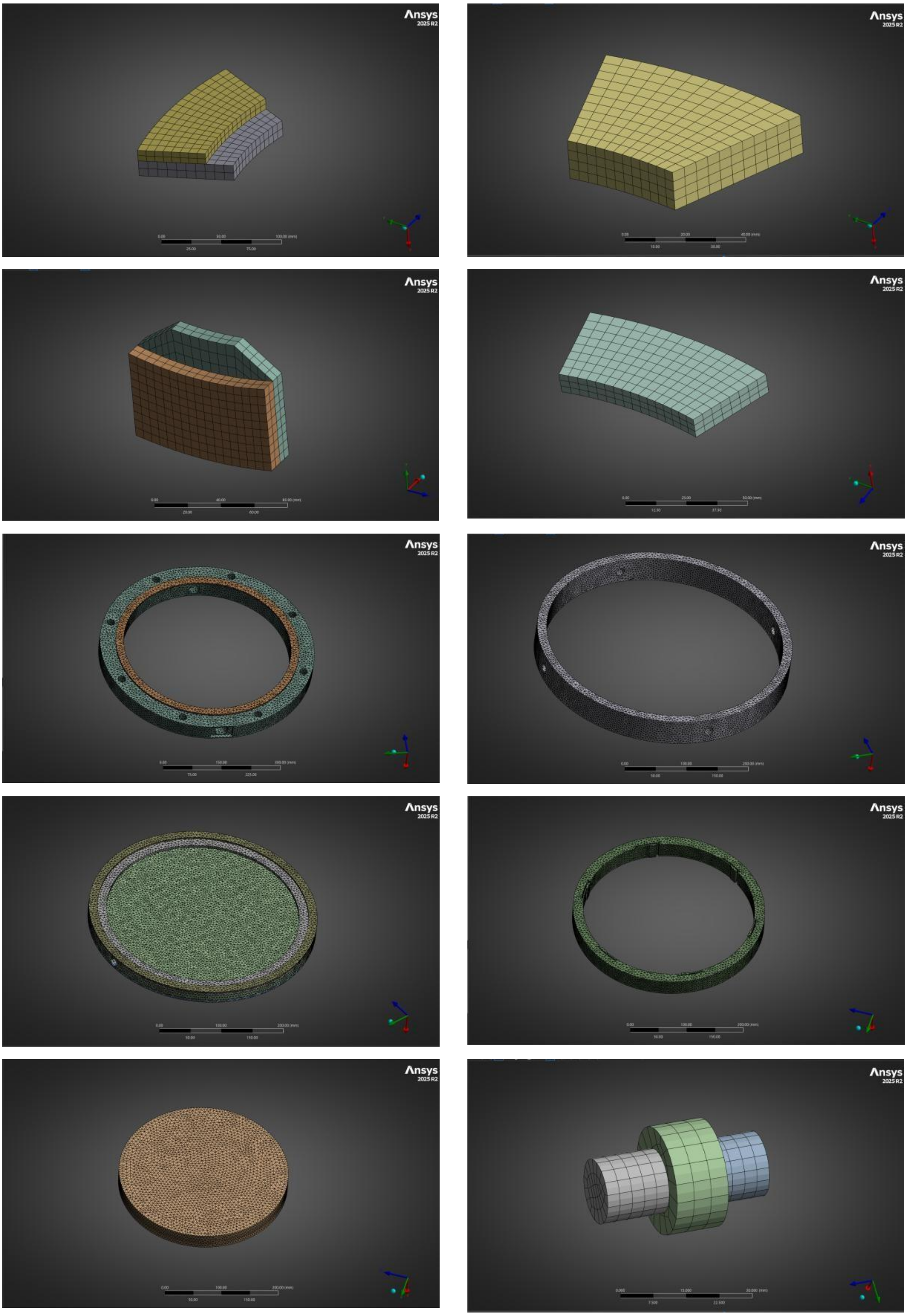


Figure 6*: Meshed views of individual assembly components.*

2.3 Mesh Quality

Mesh quality was verified using standard element-quality metrics (element quality, skewness, and aspect ratio) to confirm the mesh is suitable for the modal and random-vibration solutions.

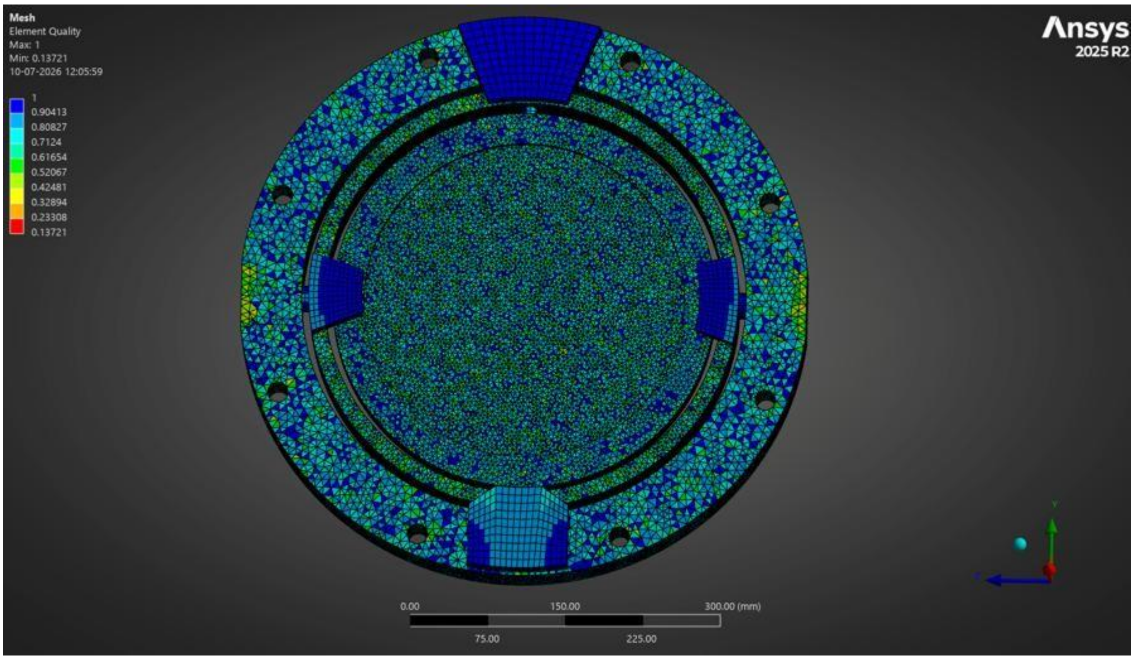


Figure 7*: Element quality distribution across the meshed assembly.*

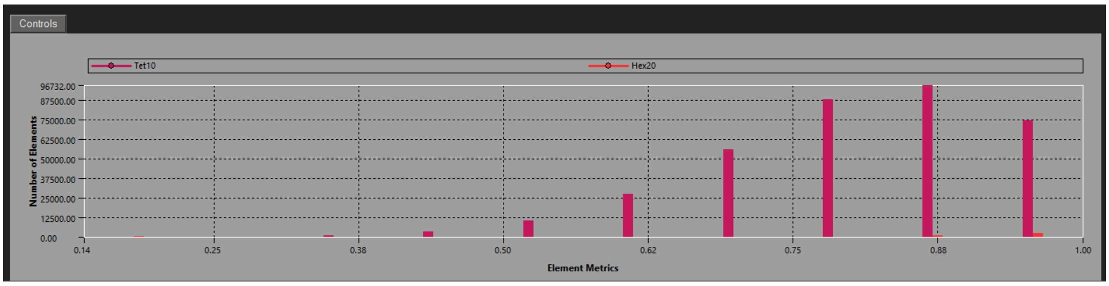


Figure 8*: Element quality — summary statistics.*

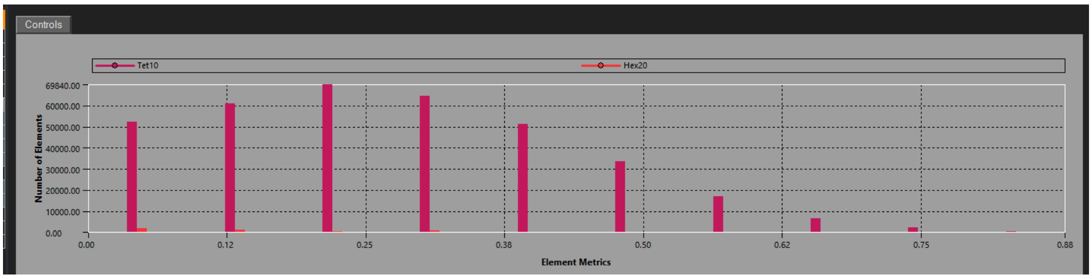


Figure 9*: Element skewness — distribution and summary statistics.*

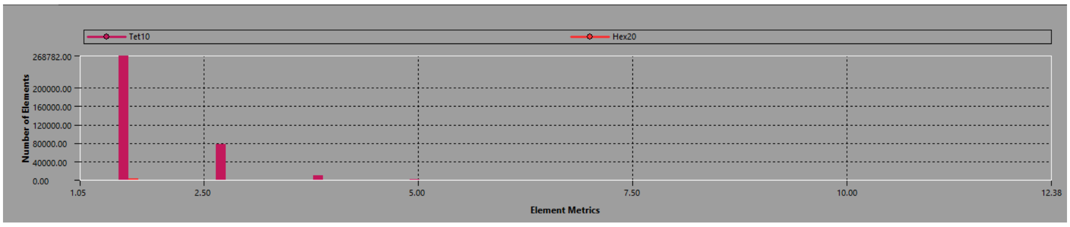


Figure 10*: Element aspect ratio — distribution and summary statistics.*

> ***Note:*** *Numerical mesh-quality metrics (minimum/maximum/average values) shown in the statistics screenshots above should be cross-checked against the analyst's mesh-quality acceptance criteria before the model is issued as final.*

2.4 Contact Definitions

All mating interfaces in the assembly were represented using bonded contacts, appropriate for a fastened/adhesively or mechanically fixed assembly where relative sliding is not expected. The dowel/locating pins ("PIN") fix the angular and radial position of the pads and outer ring relative to the base plate.

Table 2*: Contact definitions between assembly components.*

S. No.	Component A	Component B	Contact Type
1	FLANGE (5)	MICROMETER_MOUNTING_PAD_1 (9)	Bonded
2	FLANGE (5)	MICROMETER_MOUNTING_PAD_2 (10)	Bonded
3	FLANGE (5)	BEARING_CAP (8)	Bonded
4	OUTER_SHAFT (6)	BEARING_CAP (8)	Bonded
5	OUTER_SHAFT (6)	OUTER_RING (4)	Bonded
6	INNER_SHAFT (7)	OUTER_RING (4)	Bonded
7	OUTER_RING (4)	LOCKING_PAD_1 (11)	Bonded
8	OUTER_RING (4)	LOCKING_PAD_2 (12)	Bonded
9	INNER_SHAFT (7)	BASE_PLATE (3)	Bonded
10	BASE_PLATE (3)	FIXTURE_MOUNT (2)	Bonded
11	FIXTURE_MOUNT (2)	MIRROR (1)	Bonded
12	LOCKING_PAD_1 (11)	PIN	Bonded
13	LOCKING_PAD_2 (12)	PIN	Bonded
14	BASE_PLATE (3)	PIN	Bonded
15	BASE_PLATE (3)	PIN	Bonded
16	MICROMETER_MOUNTING_PAD_1 (9)	PIN	Bonded
17	MICROMETER_MOUNTING_PAD_2 (10)	PIN	Bonded
18	OUTER_RING (4)	PIN	Bonded
19	OUTER_RING (4)	PIN	Bonded

3. Modal Analysis

3.1 Methodology

- Import the CAD geometry into the FEA software.
- Clean and simplify the geometry by removing unnecessary features.
- Assign material properties (Young's modulus, Poisson's ratio, and density).
- Define contacts and connections between all components.
- Apply appropriate boundary conditions and supports.
- Generate a suitable finite element mesh and perform mesh quality checks.
- Set the required number of vibration modes (first 80 modes).
- Solve the modal analysis to obtain natural frequencies and corresponding mode shapes.
- Review the deformation patterns and mode shapes for each mode.
- Evaluate the effective mass participation factors in the X, Y, and Z directions.
- Verify that the cumulative effective mass is within the acceptable range (typically ≥ 80–90%).
- Interpret the modal results and use them for further dynamic analyses such as harmonic, random vibration, or response spectrum analysis.

3.2 Target Criteria

Modal Acceptance Target

- First (fundamental) mode frequency f ≥ 300 Hz.
- Cumulative modal effective mass ≥ 80% in each of the X, Y, and Z directions, captured within the solved mode set (first 80 modes).

3.3 Results

The first 80 modes were extracted. The frequency and effective-mass participation of the first ten modes — which capture the majority of the assembly's effective mass — are summarised in Table 3; the cumulative effective mass reaches 93.68% (X), 84.03% (Y), and 83.77% (Z) within this subset, satisfying the ≥ 80% target in all three directions.

Table 3: *Natural frequencies and modal effective mass participation — first 10 of 80 extracted modes.*

Mode	Freq. (Hz)	X Eff. Mass	X Ratio (%)	Y Eff. Mass	Y Ratio (%)	Z Eff. Mass	Z Ratio (%)
1	311.25	1.353	4.52	0.0000013	0.00	6.901	23.06
2	400.73	5.418	18.10	9.555	31.92	0.620	2.07
3	401.96	8.299	27.73	6.154	20.56	1.007	3.36
4	465.44	0.0000770	0.00	0.3286	1.10	0.000358	0.00
5	523.51	0.4268	1.43	0.000384	0.00	6.850	22.89
6	674.63	0.00546	0.02	0.1343	0.45	0.00163	0.01
7	738.29	0.000229	0.00	0.01087	0.04	0.000822	0.00
8	959.83	0.0804	0.27	0.0000874	0.00	0.000277	0.00
9	1104.0	0.00143	0.00	0.01682	0.06	0.0000877	0.00
10	1120.4	0.000258	0.00	0.1869	0.62	0.000241	0.00
Σ (1–80)	**—**	**28.04**	**93.68**	**25.15**	**84.03**	**25.07**	**83.77**

The fundamental (first) mode occurs at 311.25 Hz, satisfying the f ≥ 300 Hz target with a margin of approximately 3.8%. Modes 2 and 3 (≈ 401 Hz) contribute the largest share of X- and Y-direction effective mass, while modes 1 and 5 dominate the Z-direction response.

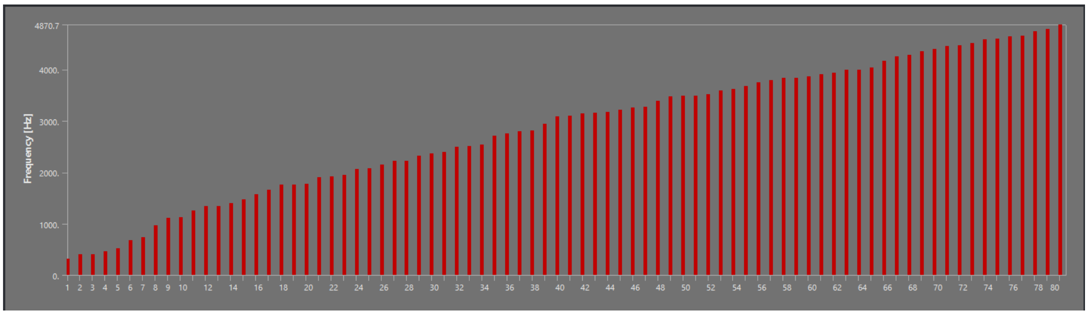


Figure 11*: Natural frequency vs. mode number for the first extracted modes.*

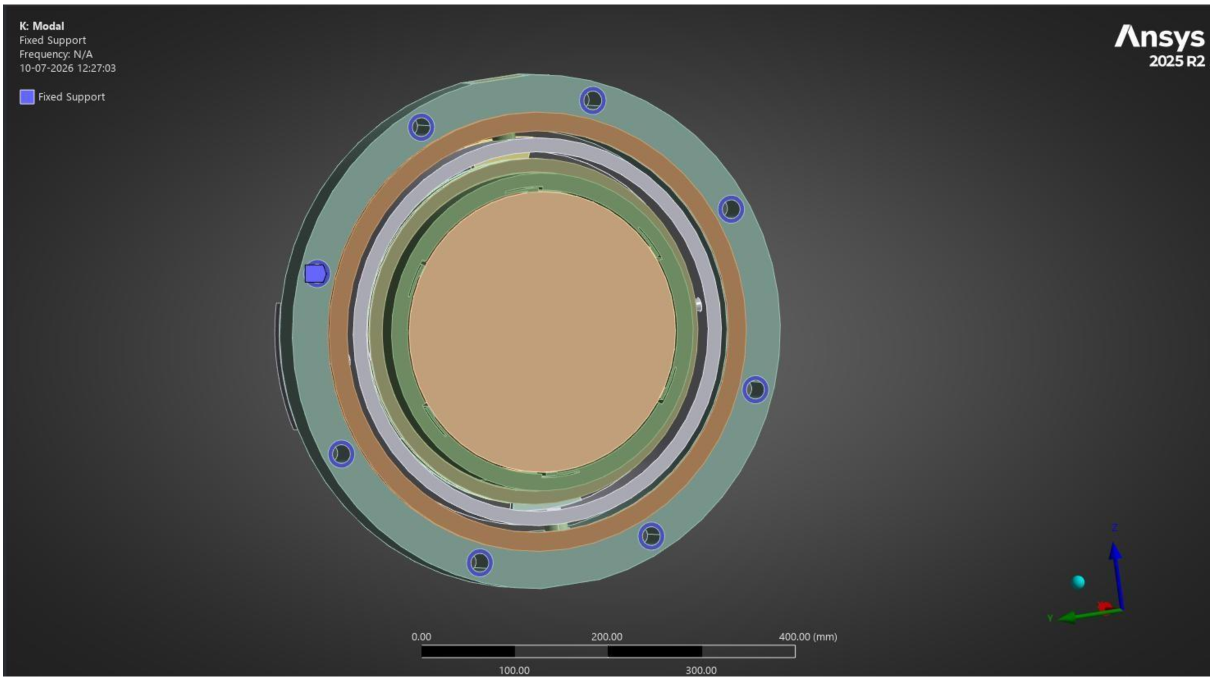
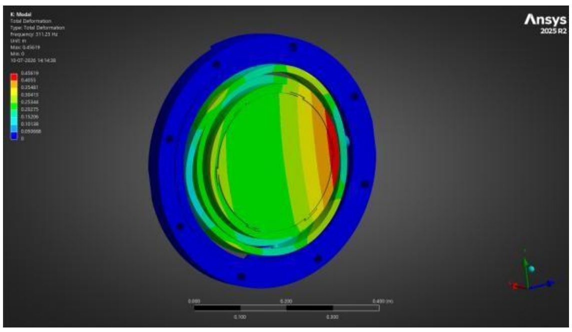


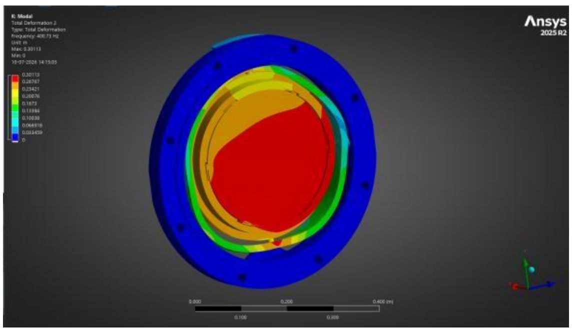
Figure 12*: Cumulative modal effective mass fraction vs. mode number.*

3.3.1 Representative Mode Shapes

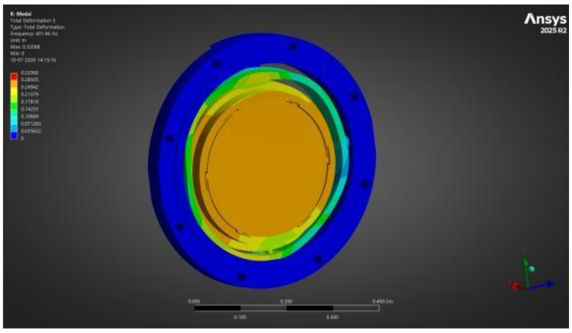
Deformation contour plots for the dominant lower-order modes are shown below; these correspond to the modes carrying the largest effective mass fractions in Table 3.



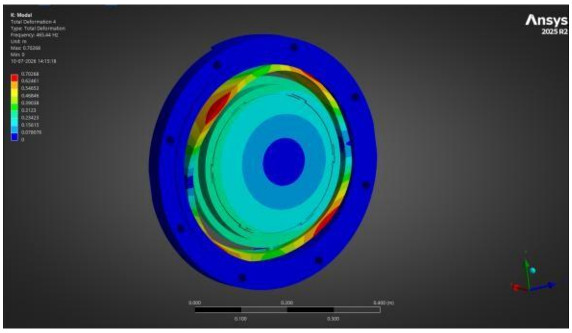
Mode 1 — 311.25 Hz



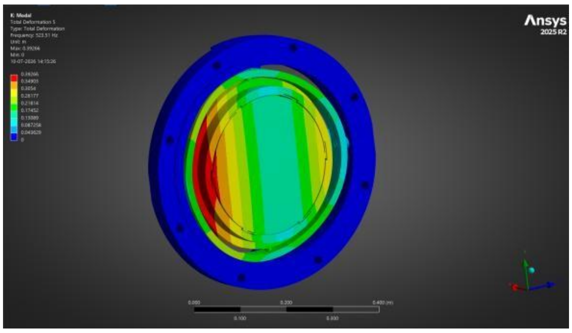
Mode 2 — 400.73 Hz



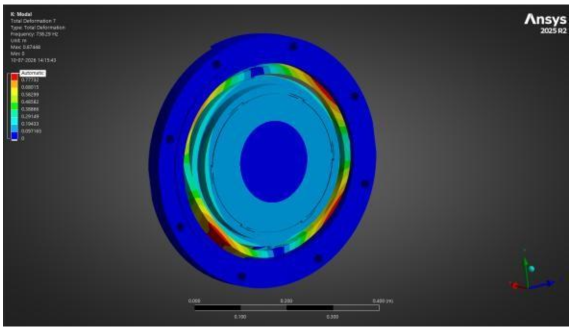
Mode 3 — 401.96 Hz



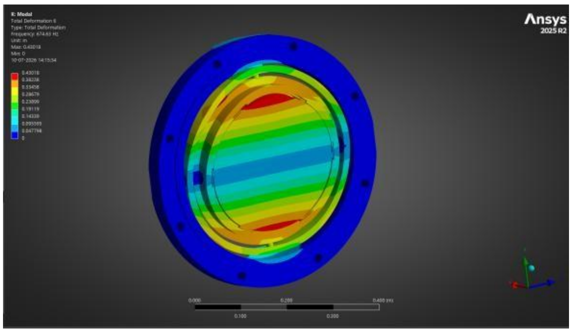
Mode 4 — 465.44 Hz



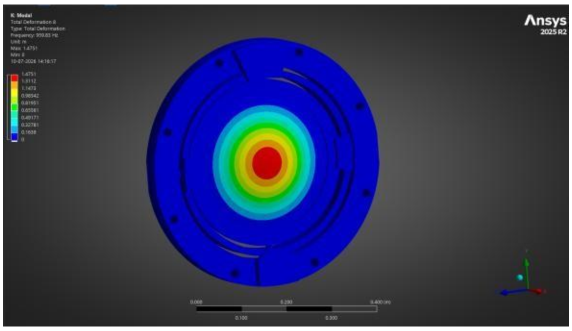
Mode 5 — 523.51 Hz



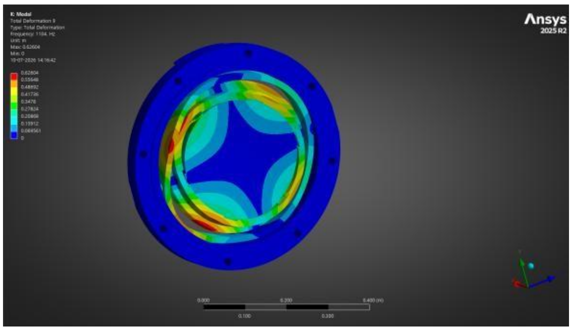
Mode 7 — 738.29 Hz



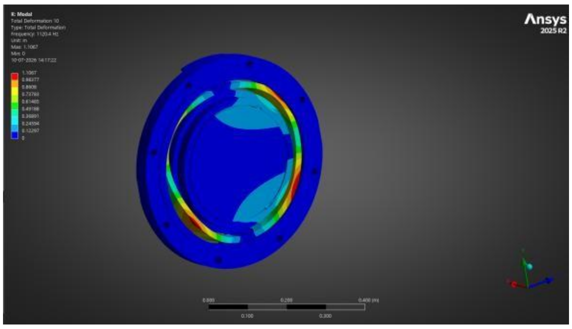
Mode 6 — 674.63 Hz



Mode 8 — 959.83 Hz



Mode 9 — 1104.0 Hz



Mode 10 — 1120.4 Hz

Figure 13*: Representative mode-shape deformation contours for the dominant lower-order modes.*

> ***Note:*** *This section summarises the methodology and results as provided by the analyst's ANSYS output. Material properties, mesh density, and exact boundary/support conditions used in the model should be verified against the source ANSYS project before this document is issued as a formal analysis report.*

4. Random Vibration (PSD) Analysis

Mirror Assembly — PSD Base Excitation Study (ANSYS Mechanical 2025 R2)

4.1 Objective

The objective of this analysis is to evaluate the dynamic structural response of the mirror assembly under a random (PSD) base excitation representative of the specified vibration environment, and to confirm that the resulting mirror tilt (clocking / tip-tilt error induced by structural deformation) remains within the optical alignment budget of 5 microradians (μrad), evaluated at the 3-sigma (99.73% probability) response level.

4.2 Analysis Method Overview

Random vibration analysis in ANSYS Mechanical is a linear, spectral method that propagates an input Power Spectral Density (PSD) through the structure's modal characteristics to obtain statistical (RMS / 1-sigma) response quantities, which are then scaled to the desired probability level (1σ, 2σ, or 3σ). The workflow used for this study is summarised below:

- Modal analysis — extraction of natural frequencies and mode shapes of the assembly over a frequency range that comfortably exceeds the highest frequency of the input PSD (typically 1.5–2× the input cut-off), ensuring adequate modal mass participation is captured.
- Random Vibration (PSD) analysis — linked to the modal solution; the input PSD acceleration spectrum is applied as a base excitation and combined with the extracted mode shapes using mode-superposition to compute the statistical response (displacement, stress, and derived quantities).
- Post-processing — equivalent (von-Mises) stress and a user-defined tilt result are evaluated at the 3-sigma probability level to assess structural adequacy and optical performance against the target.

4.3 Input Definition — PSD Acceleration Spectrum

The input excitation was defined as a base-acceleration PSD applied at the mounting/support interface of the assembly, as illustrated in Figure 14. The breakpoints used to define the PSD curve are listed in Table 4 and shown graphically in Figure 15.

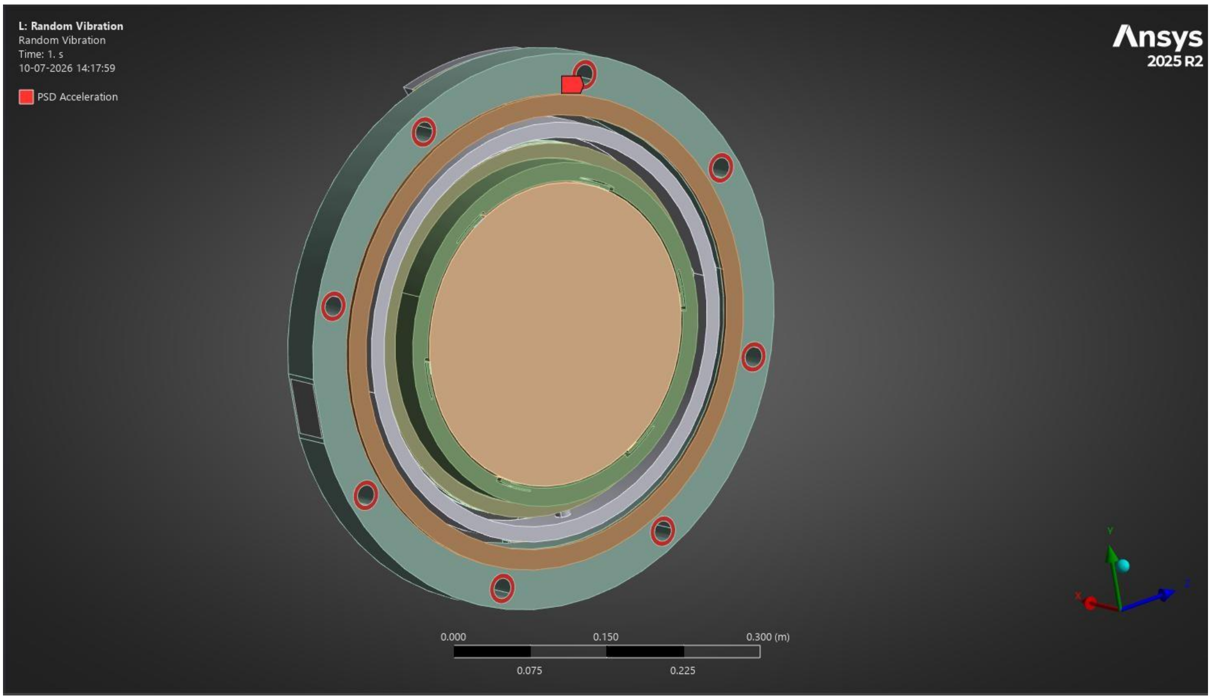


Figure 14*: Random vibration excitation applied at the fixed support.*

Table 4*: Input PSD breakpoints.*

Frequency (Hz)	PSD Acceleration ((m/s²)²/Hz)
20.00	2.0
50.00	2.0
107.72	0.73681
232.08	0.27144
500.00	0.1

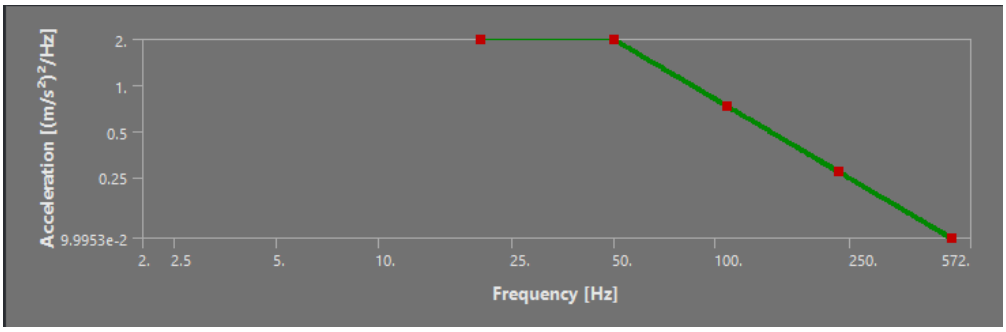


Figure 15*: Input acceleration PSD vs. frequency (log-log).*

The spectrum is flat at 2.0 (m/s²)²/Hz from 20–50 Hz and then rolls off at a constant slope to 0.1 (m/s²)²/Hz at 500 Hz, consistent with a typical qualification-level random vibration profile.

4.4 Finite Element Model and Boundary Conditions

The following modelling approach was adopted for the study; assembly-specific details (materials, mesh density, and exact fixity locations) should be confirmed against the analysis model documentation:

- Geometry: full mirror mounting assembly (mirror, fixture mount, base plate, outer ring, flange, shafts, bearing cap, mounting/locking pads, and fasteners) with bonded contacts defined at all mating interfaces per the contact schedule.
- Supports: fixed / remote displacement support applied at the mounting interface representing the base excitation attachment point.
- Loading: base PSD acceleration applied in the direction of primary sensitivity (through the modal analysis boundary condition set), consistent with the input spectrum in Section 4.3.
- Damping: a representative structural damping ratio (commonly 2% critical, unless test-derived values are available) applied uniformly across the modes contributing to the response.
- Solver output: results requested at the 3-sigma (99.73% probability) level, consistent with standard aerospace/opto-mechanical qualification practice.

4.5 Mirror Tilt Derivation (User-Defined Result)

Optical tilt is not a native ANSYS output and was derived using a user-defined result expressing the small-angle rotation of the mirror surface. The expression UY/LOCX approximates the local tilt angle (in radians) by dividing the out-of-plane displacement (UY) by the local radial coordinate (LOCX) at each node, which is valid under the small-angle assumption ($\theta \approx \tan\theta = \Delta y/x$) applicable to the sub-micron deformations expected here. This result was evaluated at the same 3-sigma probability level as the stress solution so that stress and tilt are reported on a consistent statistical basis.

4.6 Results

4.6.1 Equivalent Stress

Figure 16 shows the 3-sigma equivalent (von-Mises) stress distribution across the assembly. The maximum stress of 183.61 MPa occurs locally near a bonded/contact interface, with the bulk of the structure well below this value (minimum 902 Pa).

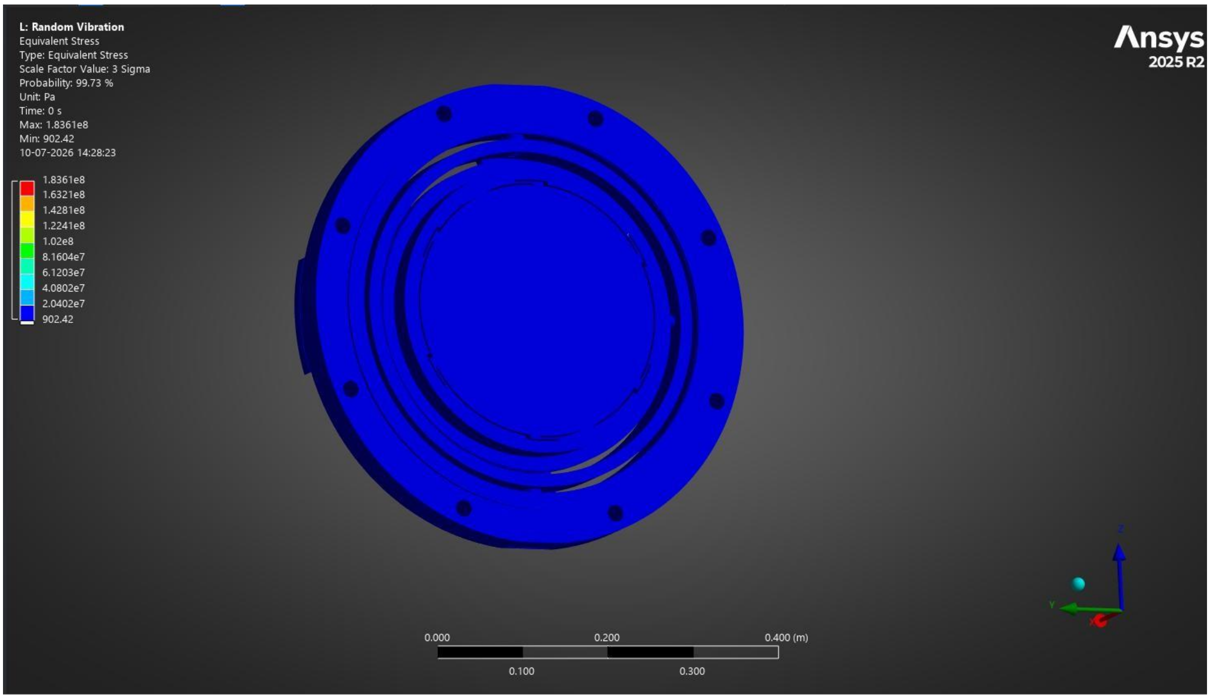
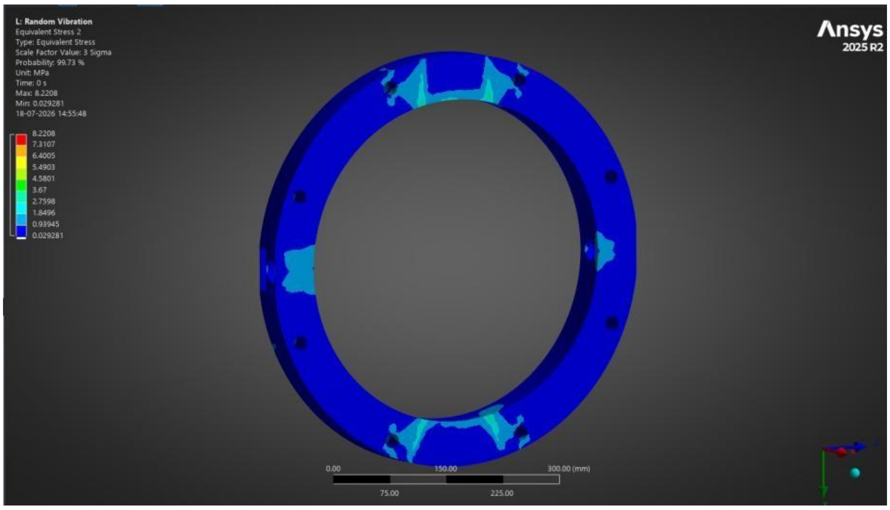


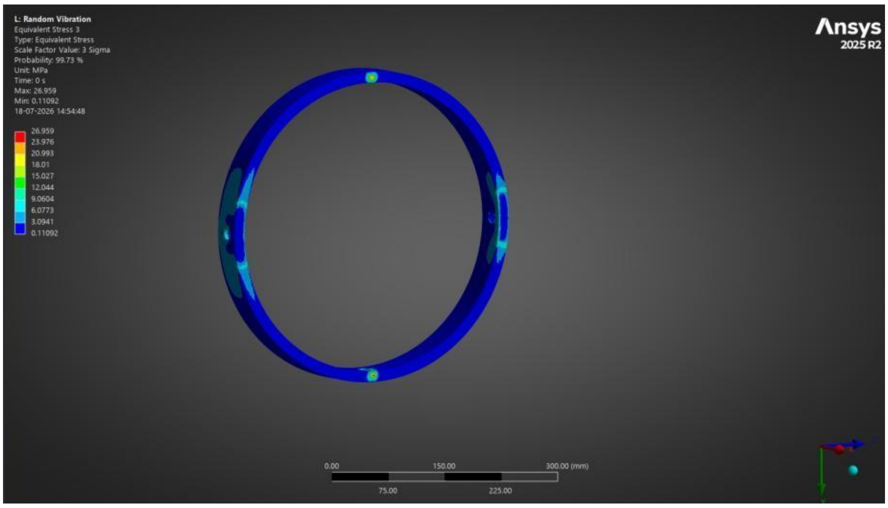
Figure 16: *3σ equivalent stress distribution (Max 1.8361 × 10^8 Pa, Min 902.42 Pa).*

Individual Component Stress Distributions

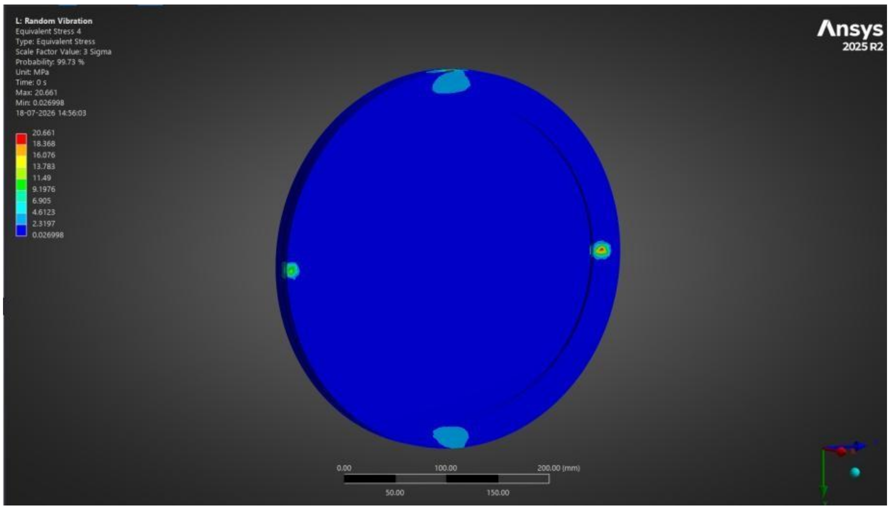
To supplement the assembly-level 3σ equivalent stress result above, the random-vibration solution was also post-processed on a per-component basis. Figures below show the 3σ equivalent (von-Mises) stress distribution for each individual load-bearing component, followed by a summary of the corresponding maximum and minimum values (Table below).



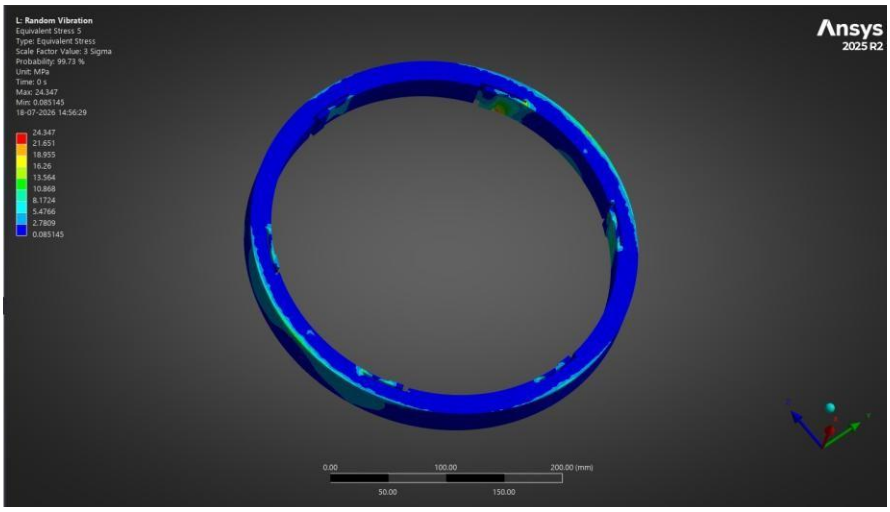
Item 2 — FIXTURE_MOUNT: Max = 8.2208 MPa, Min = 0.029281 MPa



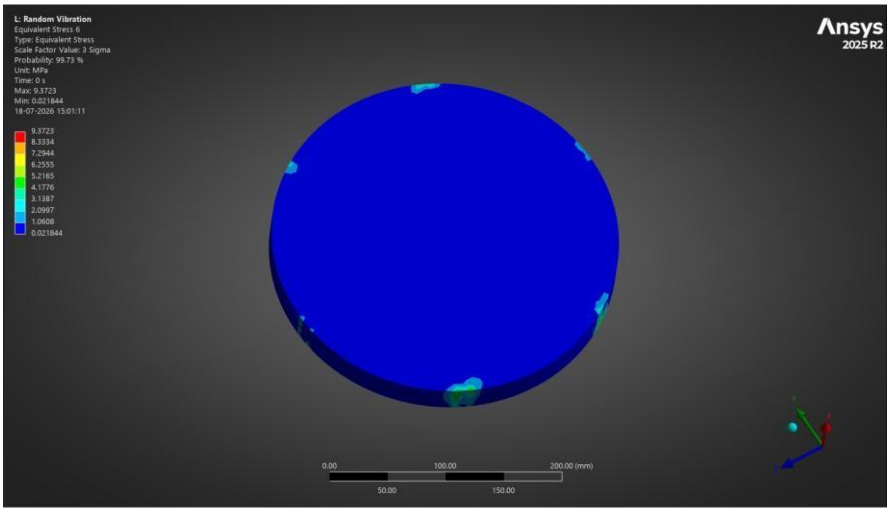
Item 3 — BASE_PLATE: Max = 26.959 MPa, Min = 0.11092 MPa



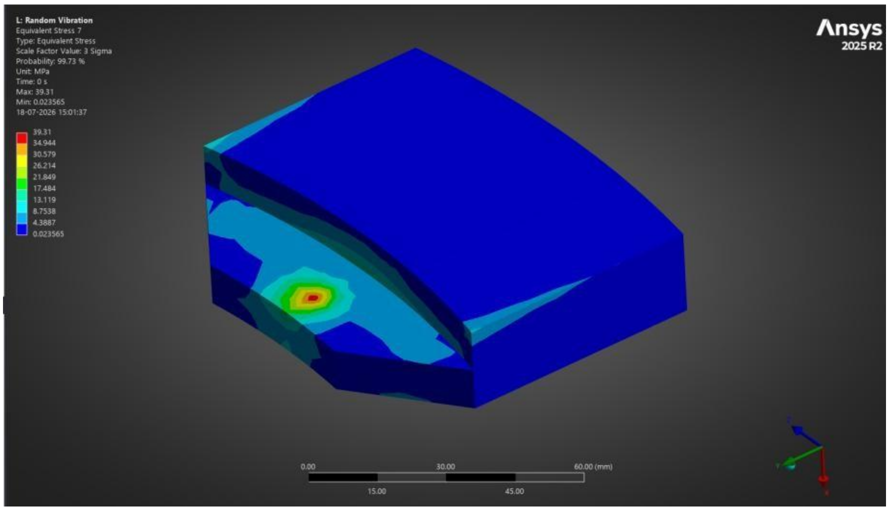
Item 4 — OUTER_RING: Max = 20.661 MPa, Min = 0.026998 MPa



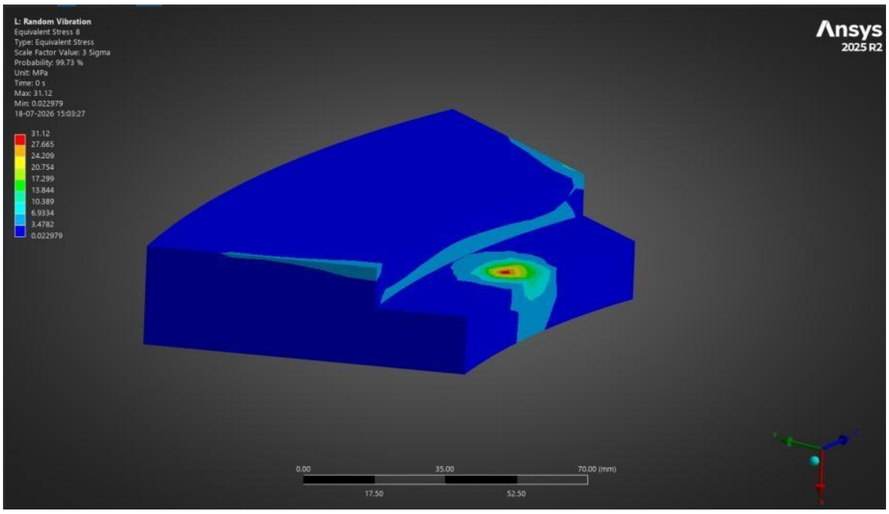
Item 5 — FLANGE: Max = 24.347 MPa, Min = 0.085145 MPa



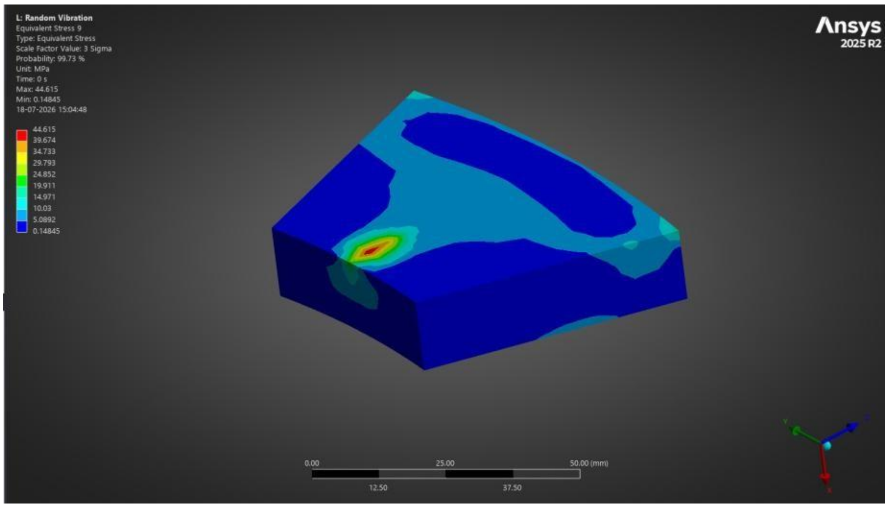
Item 6 — OUTER_SHAFT: Max = 9.3723 MPa, Min = 0.021844 MPa



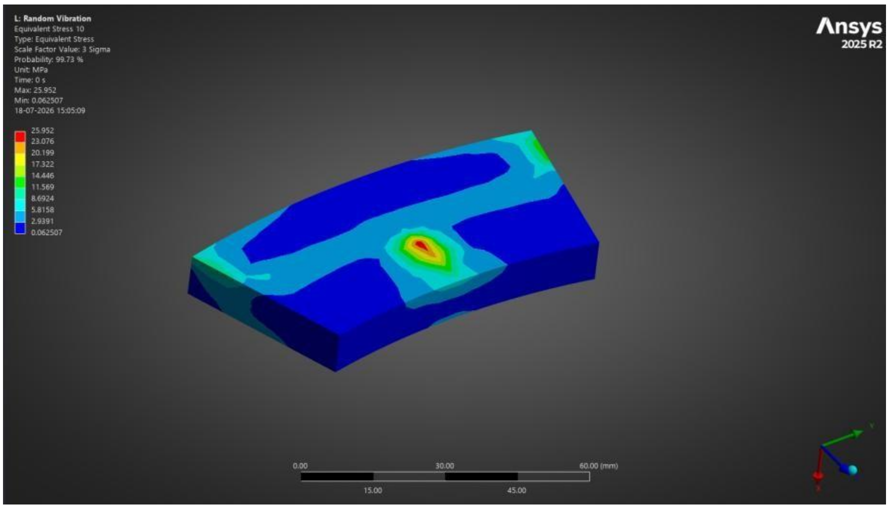
Item 7 — INNER_SHAFT: Max = 39.31 MPa, Min = 0.023565 MPa



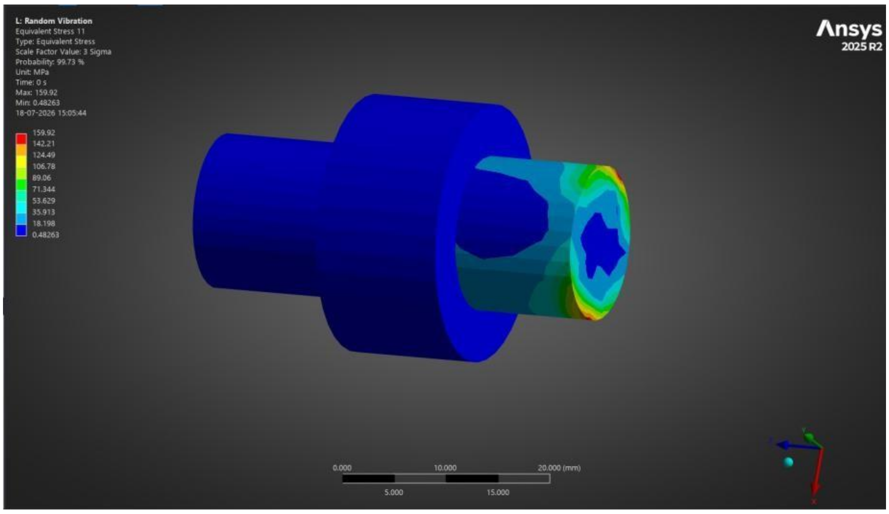
Item 8 — BEARING_CAP: Max = 31.12 MPa, Min = 0.022979 MPa



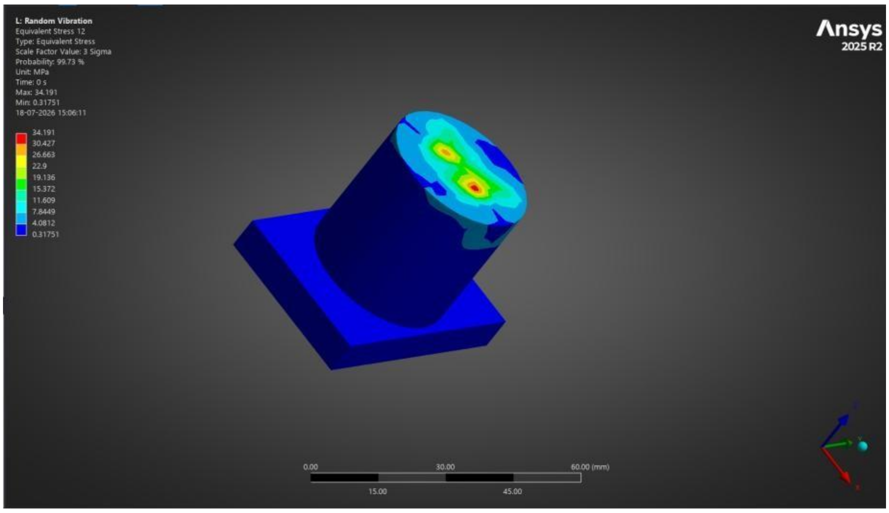
Item 9 — MICROMETER_MOUNTING_PAD_1: Max = 44.615 MPa, Min = 0.14845 MPa



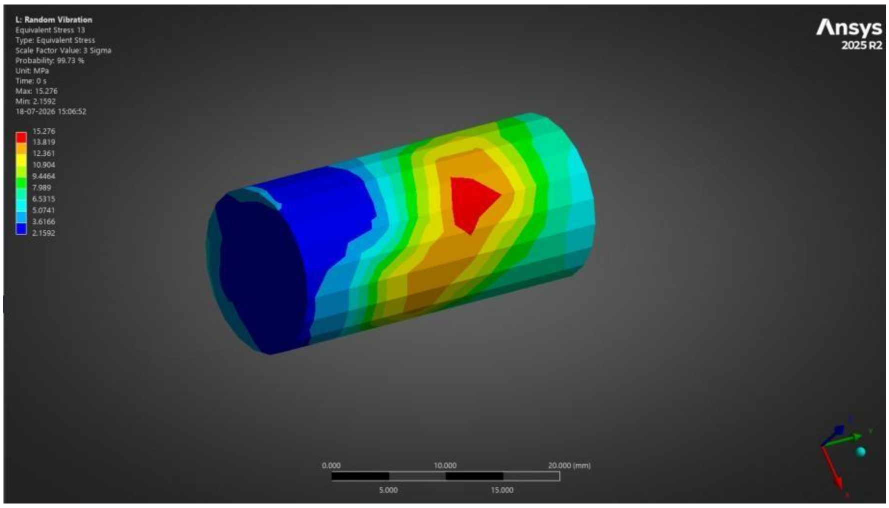
Item 10 — MICROMETER_MOUNTING_PAD_2: Max = 25.952 MPa, Min = 0.062507 MPa



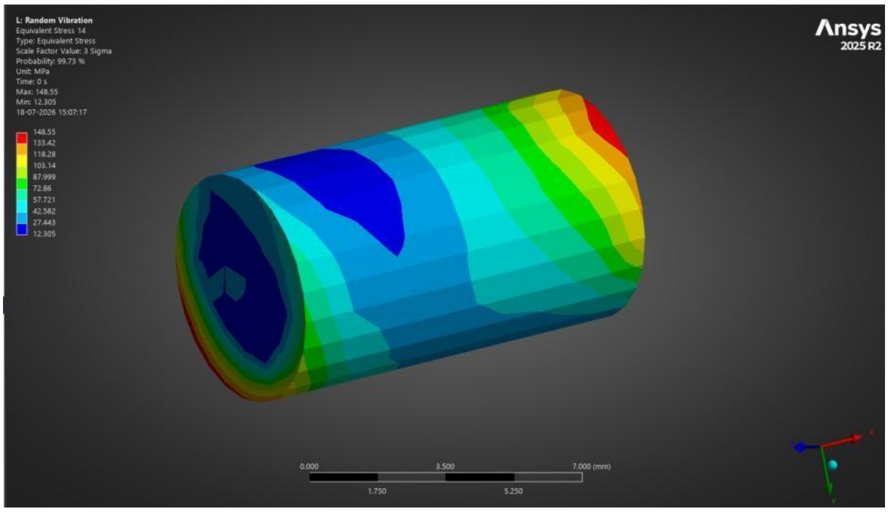
Item 11 — LOCKING_PAD_1: Max = 159.92 MPa, Min = 0.48263 MPa



Item 12 — LOCKING_PAD_2: Max = 34.191 MPa, Min = 0.31751 MPa



Item 13 — 6202_1 (Bearing): Max = 15.276 MPa, Min = 2.1592 MPa



Item 14 — LOCK SCREW: Max = 148.55 MPa, Min = 12.305 MPa

Component Stress Summary

Item No.	Component	Max. Equiv. Stress (MPa)	Min. Equiv. Stress (MPa)
2	FIXTURE_MOUNT	8.2208	0.029281
3	BASE_PLATE	26.959	0.11092
4	OUTER_RING	20.661	0.026998
5	FLANGE	24.347	0.085145
6	OUTER_SHAFT	9.3723	0.021844
7	INNER_SHAFT	39.31	0.023565
8	BEARING_CAP	31.12	0.022979
9	MICROMETER_MOUNTING_PAD_1	44.615	0.14845
10	MICROMETER_MOUNTING_PAD_2	25.952	0.062507
11	LOCKING_PAD_1	159.92	0.48263
12	LOCKING_PAD_2	34.191	0.31751
13	6202_1 (Bearing)	15.276	2.1592
14	LOCK SCREW	148.55	12.305

Note: component numbering follows the item numbers assigned in the Bill of Materials (Table 1); stresses are reported at the 3σ (99.73% probability) level, consistent with the assembly-level random vibration result above.

4.6.2 Response PSD

Figure 17 shows the acceleration response PSD extracted at a representative point on the assembly, plotted over the 20–500 Hz analysis range. Two resonant amplifications are visible, with the dominant peak (~299 $(m/s^2)^2/Hz$) occurring near the upper end of the frequency range, indicating the structural modes most strongly excited by the input spectrum.

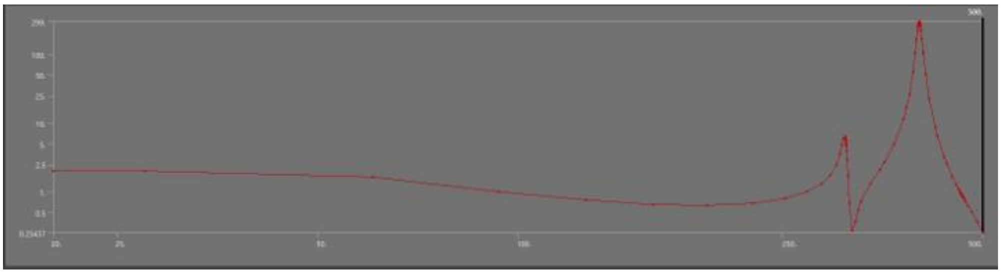


Figure 17*: Output acceleration response PSD (log-log).*

4.6.3 Mirror Tilt

Figure 18 shows the spatial distribution of the derived tilt result (UY/LOCX) across the mirror face at the 3-sigma level. The tilt varies smoothly across the aperture, with a maximum of 3.5673 × 10^{-6} rad (3.567 µrad) and a minimum of 2.5377 × 10^{-6} rad (2.538 µrad).

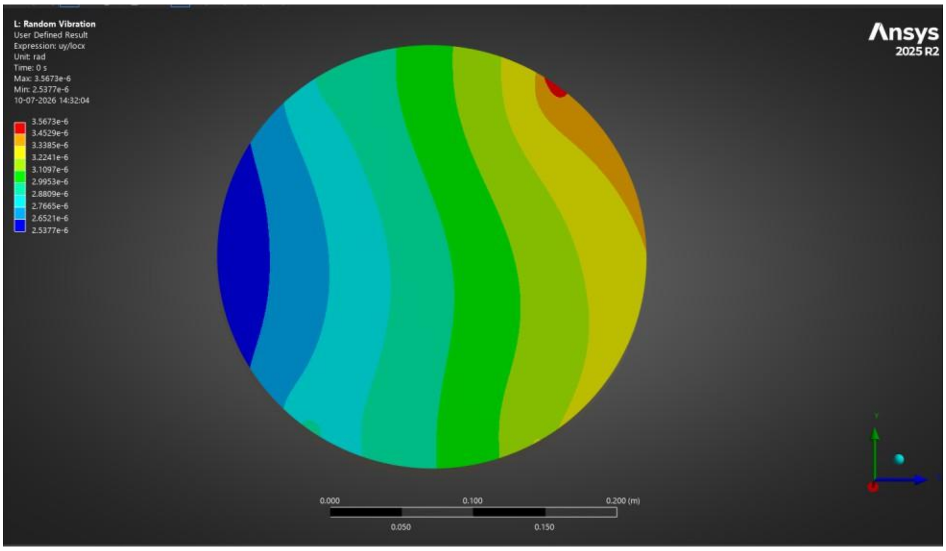


Figure 18*: 3σ mirror tilt distribution, UY/LOCX (rad).*

4.7 Summary

Table 5*: Random vibration response summary against target criteria.*

Response Quantity	3σ Value	Target	Status
Max. equivalent (von-Mises) stress	183.61 MPa	Reference only	For information
Min. equivalent stress	0.90 kPa	—	—
Max. mirror tilt (UY/LOCX)	3.567 µrad	< 5 µrad	PASS
Min. mirror tilt (UY/LOCX)	2.538 µrad	< 5 µrad	PASS

4.8 Conclusion

Under the specified input PSD acceleration profile, the assembly's maximum predicted mirror tilt at the 3-sigma (99.73% probability) response level is 3.567 µrad, which is within the 5 µrad optical alignment target, providing a margin of approximately 29%. The peak equivalent stress of 183.61 MPa should be compared against the applicable material allowable (with appropriate factor of safety) to confirm structural adequacy; this was not specified as part of the current acceptance criteria.

> ***Note:*** *This section summarises the methodology and results as provided by the analyst's ANSYS output screenshots. Model-specific inputs (material properties, damping ratio, exact boundary conditions, and mesh statistics) should be verified against the source ANSYS project before this document is issued as a formal analysis report.*

5. Overall Conclusion and Recommendations

5.1 Summary of Findings

- Modal: the fundamental mode occurs at 311.25 Hz (≥ 300 Hz target met, ~3.8% margin); cumulative modal effective mass within the first 10 (of 80) modes reaches 93.68% / 84.03% / 83.77% in X / Y / Z, satisfying the ≥ 80% target in all directions.
- Random Vibration: maximum mirror tilt at the 3σ level is 3.567 µrad, within the 5 µrad optical alignment target (~29% margin); peak equivalent stress is 183.61 MPa, to be checked against material allowables.